```
//! Enums
    /*
        ? it is used :
            ~ it is a user-defined data type that consists of a set of named integral constants,
            ~  called enumerators.
            ! An Enumeration is a distinct type whose value is restricted to a range of values
            # it is represent a collection of related values that can be used to represent a
            # specific set of options or states in a program.
            # it is a way to give more meaningful names to a set of related values, making
            # the code more readable and maintainable.
        ? syntax:
            enum EnumName {
                Enumerator1,
                Enumerator2,
                Enumerator3,
                ...
            };

        ! note :
            ? by default, the first enumerator is assigned the value 0, and each subsequent enumerator is assigned
            ? a value that is one greater than the previous enumerator.
            ? However, you can also explicitly assign values to the enumerators if you want to
            ? start from a different value or have specific values for each enumerator.

    */

```

```
    //~ Example of enum:
    enum Color {
        Red,     // 0
        Green,   // 1
        Blue     // 2
    };

    //~ Example of enum with assigned values:
    enum Day {
        Sunday = 1,
        Monday = 2,
        Tuesday = 3,
        Wednesday = 4,
        Thursday = 5,
        Friday = 6,
        Saturday = 7
    };
    //~ Example of enum with assigned values and auto-increment:
    enum Status {
        Pending = 1,
        InProgress, // 2
        Completed,  // 3
        Cancelled   // 4
    };

```

```cpp
//~ Example of enum with assigned values and auto-increment:
enum Direction {
    North = 100,
    East = 200,
    South = 300,
    West = 400
};
//~ Example of enum with assigned values and auto-increment:
enum Month {
    January = 1,
    February,   // 2
    March,      // 3
    April,      // 4
    May,        // 5
    June,       // 6
    July,       // 7
    August,     // 8
    September,  // 9
    October,    // 10
    November,   // 11
    December    // 12
};

```

```cpp
//how to use enum:
#include <iostream>
using namespace std;
enum Color {
    Red,
    Green,
    Blue
};

 enum Direction {
    North ,
    East,
    South,
    West
};

int main() {
    Color myColor;
    Direction myDirection;


    myColor = Color::Green; // Assigning an enumerator to a variable
    myDirection = Direction::South; // Assigning an enumerator to a variable

    cout << "The value of myColor is: " << myColor << endl; // Output: 1
    return 0;
}

```